AutoVox

Team Five Of A Kind
Members: Omar, Khalid, Andrew, Daniel, Liv

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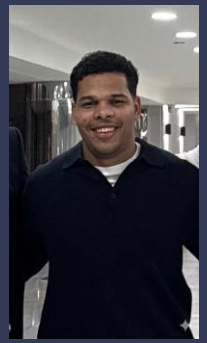
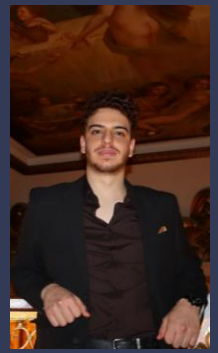
THE TEAM

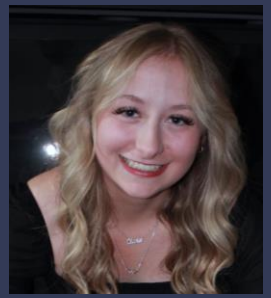


Khalid Farag

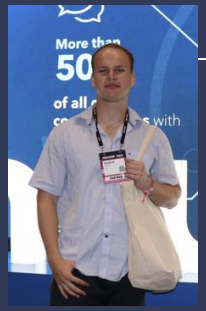
Daniel Maurer



Omar Abdelhamid



Liv Reich



Andrew Bovbel

The Problem

- Graduate students need to improve current systems for building 3D printed magnetic micro robots
- Requires a system that allows them to easily voxelize their models with sufficient metadata storage
- Unable to use prefer software like AutoCAD for building 3D models

Issues with COMSOL

Build entire model voxel-by-voxel – long, laborious & repetitive process

Extra prep for complex models – requires planning model on paper first

Unreliable software with large models where crashes can lead to lost work

PROBLEM CONTINUED

Slow, time-consuming & inefficient

Leverage existing tools

Separate the process of voxelization

Why This Matters

This workflow limits how quickly engineers can design and test advanced 3D printed objects.

Our Solution

- Convert any STL into an editable voxel model — instantly
- Layer-by-layer control of material and magnetization

What We Accomplished

The Starting Point

A multi-day voxel by voxel building on unreliable software ☹

Our Solution

We developed a system that converts STL models into editable voxel structures, enabling precise control over material and magnetic properties. ☺

Project Features

Feature 1

Import & voxelization

Feature 2

Partition & Layer Breakdown

Feature 3

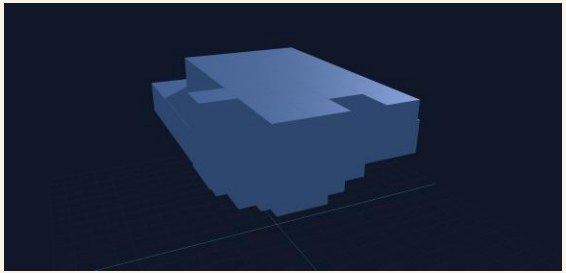
Voxel Editing

Feature 4

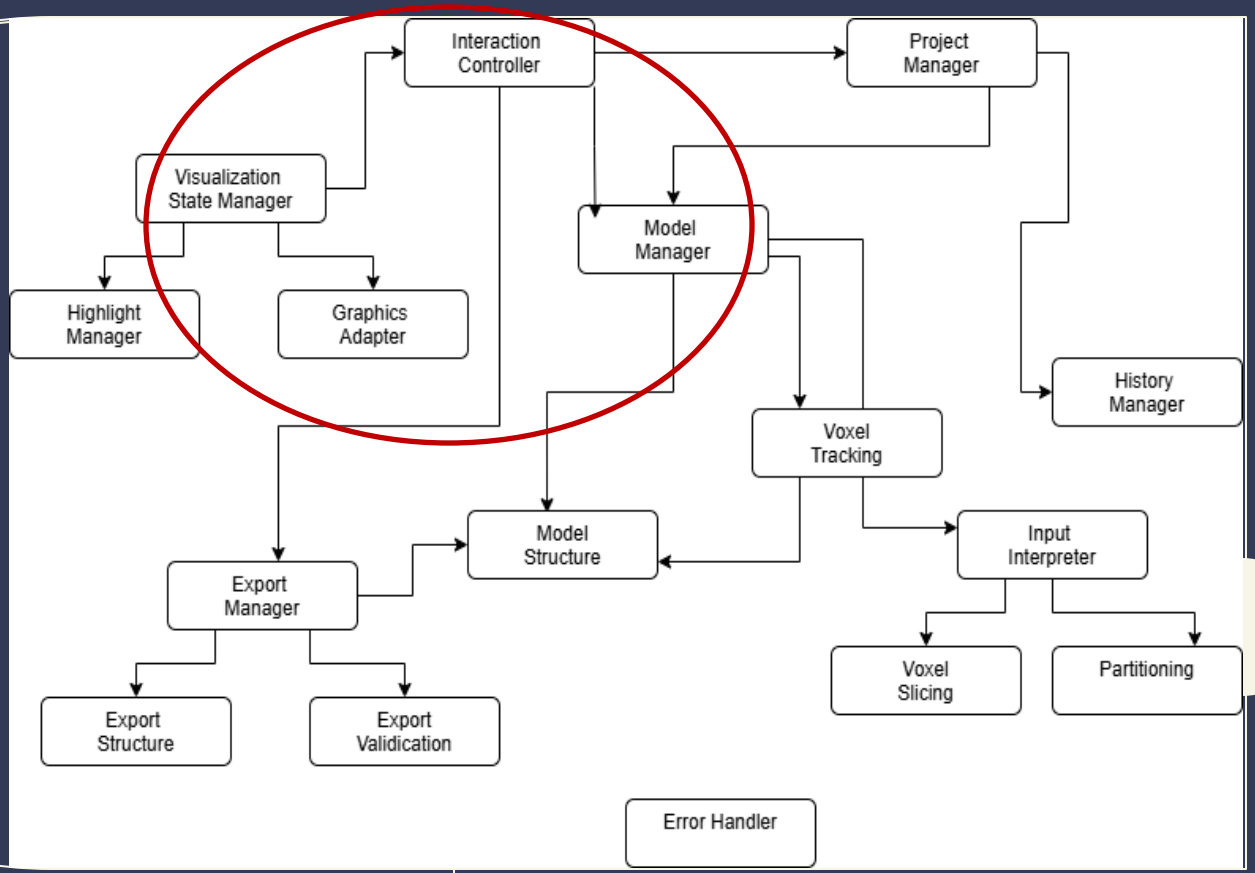
Exporting to CSV

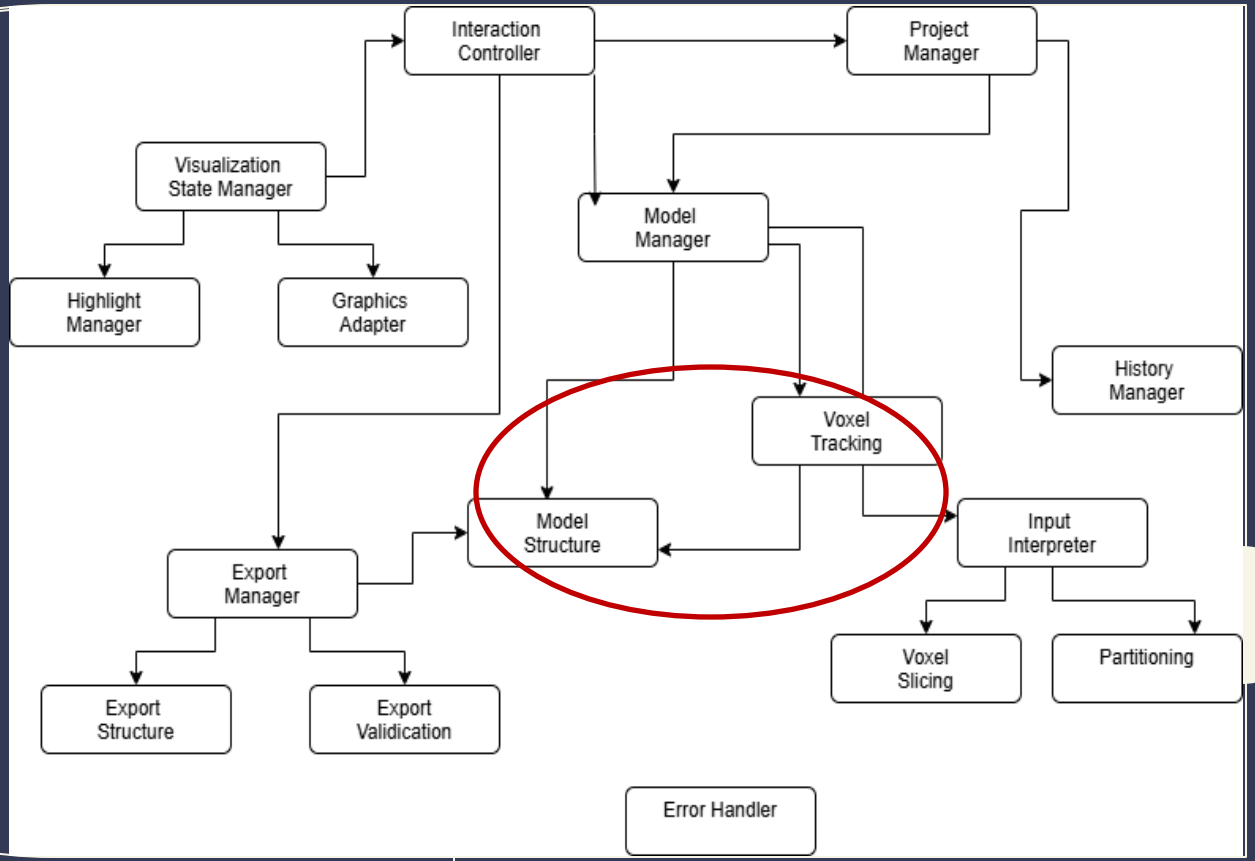
Live Demo

Let's see how it works in practice!!



System architecture


Interaction Controller
Project Manager
Visualization State Manager
Model Manager
Highlight Manager
Graphics Adapter
History Manager
Voxel Tracking
Model Structure
Input Interpreter
Export Manager
Voxel Slicing
Partitioning
Export Structure
Export Validication
Error Handler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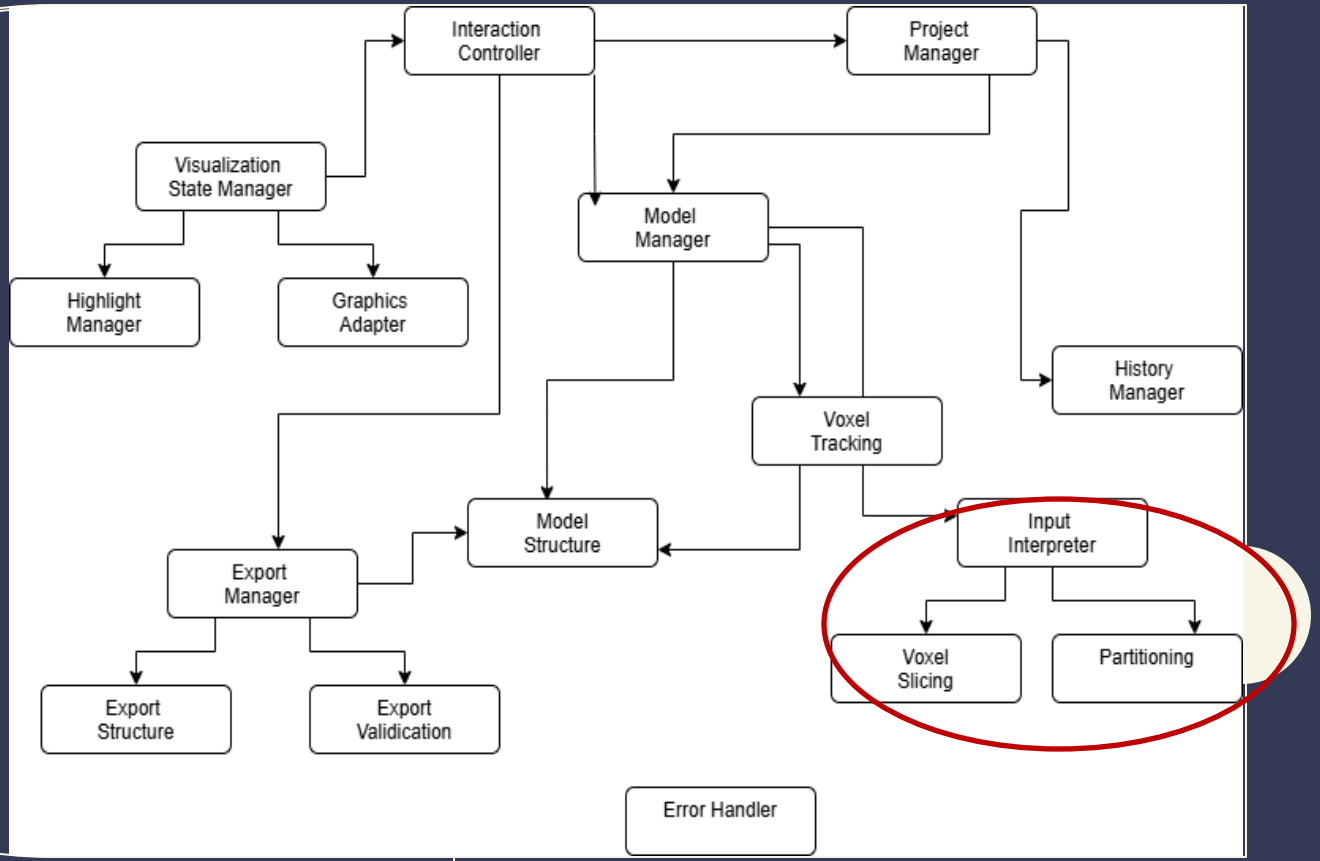
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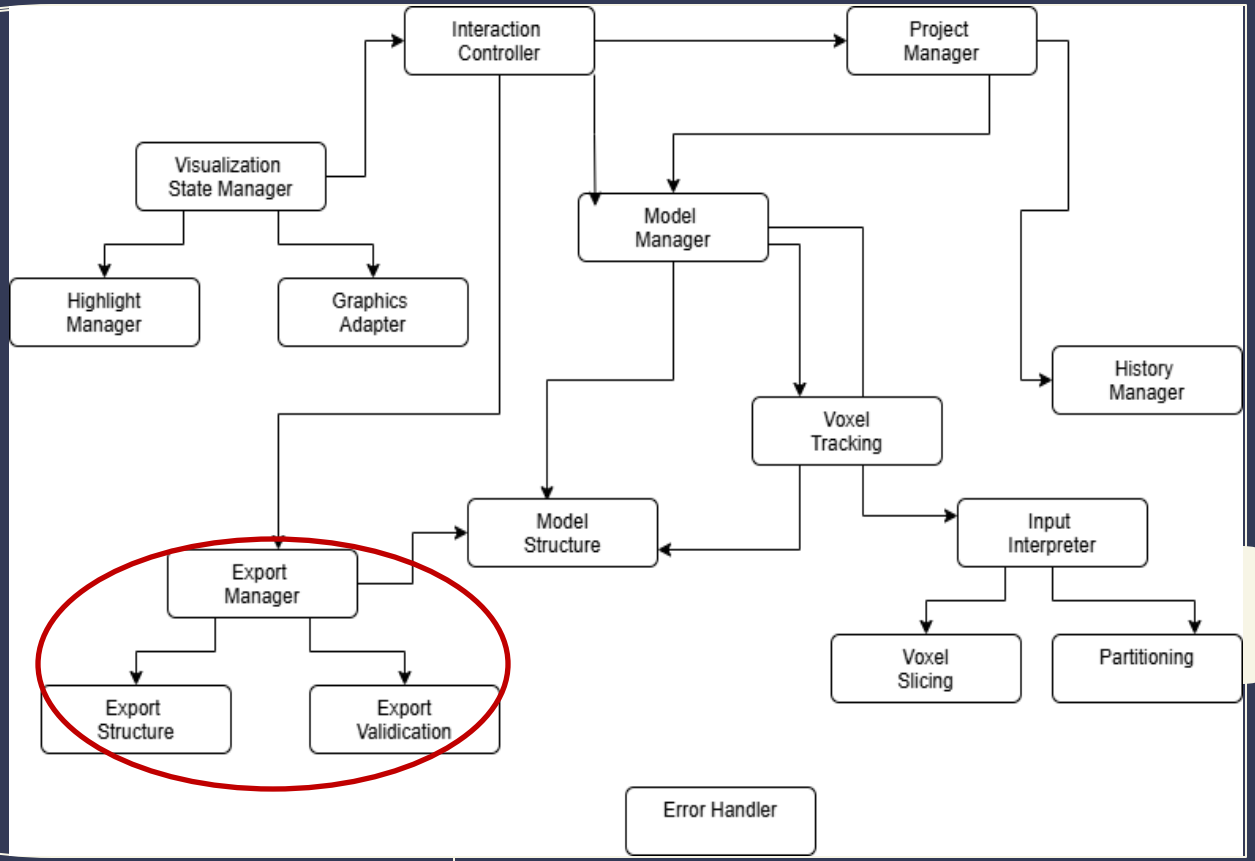
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Usability testing results

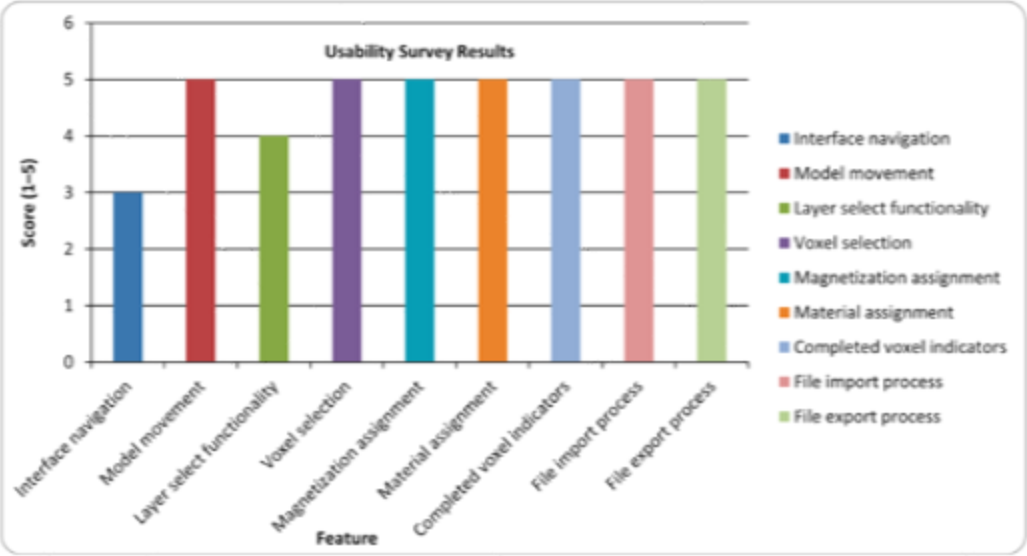
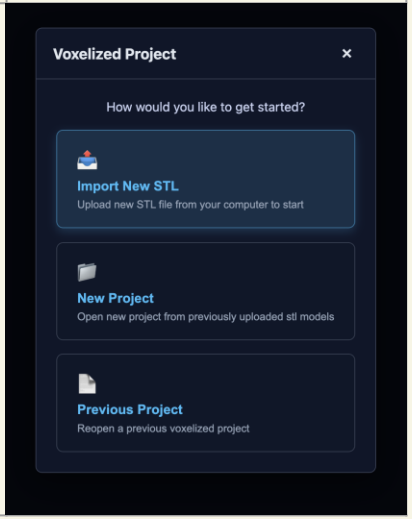


Figure 1: User Usability Survey Results

Usability testing results

Feedback Area	User Feedback	Improvement
Magnetization Visualization	Users suggested adding a visual indicator (such as an "X") on voxels whose magnetization has been modified from the default value.	

Usability testing results

Feedback Area	User Feedback	Improvement
Voxelized File Loading	Some users found the process of opening and loading voxelized files slightly more complex due to the additional interaction step required.	

Usability testing results

Feedback Area	User Feedback	Improvement
Layer Viewing Orientation	Some participants suggested that viewing layers from a top-to-bottom orientation may be more intuitive than the current right-to-left.	

Usability testing results

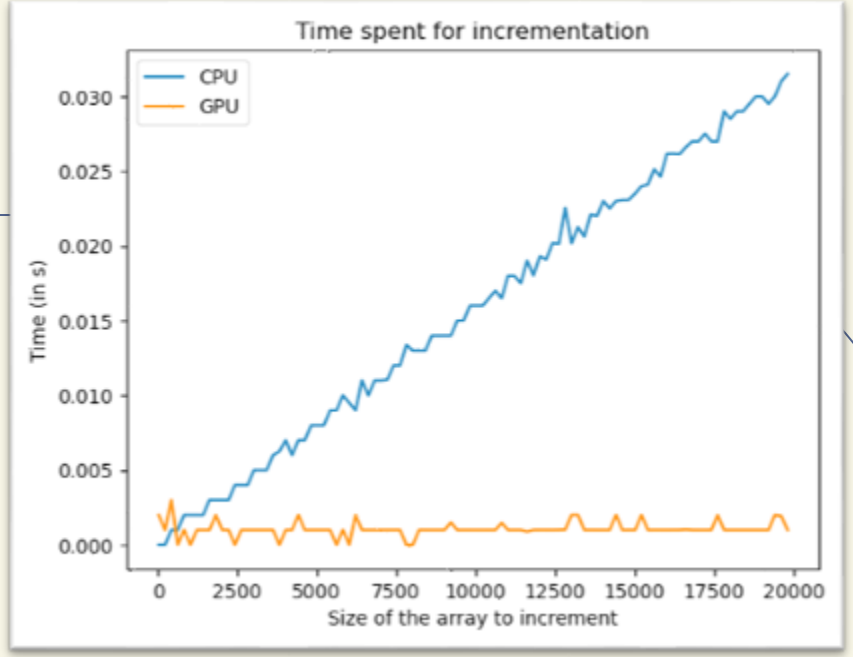
Feedback Area	User Feedback	Improvement
Partition Management	Users suggested the ability to rename partitions after they have been automatically created	

Features to implement from first test

Feedback Area	User Feedback	Implementation Plan
Partition Viewing	Users suggested that partitions should not always be centered when viewing	Allow for toggle to switch between viewing centered and showing where the partition lies in the model

Feedback from second usability test

Feedback Area	User Feedback	Implementation Plan
Drag model	Partition and layer panels block the view of the model, however after a second round, they preferred as is where you can drag the model away manually.	Inquire with the team more to see if a toggle feature would be preferred or keep as is.
Update via enter	User asked for a simple enter command whenever filling out magnetization values.	Add a simple event listener for ENTER to update the field.

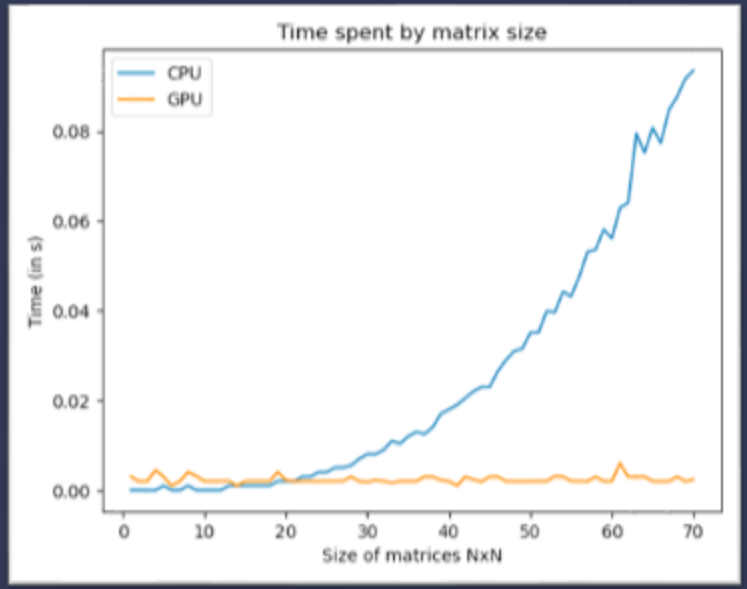


Increment array +1 (handles bigger model increments) in NumPy vs CuPy
Source: ArthurHoa (2022), GPU vs CPU (GitHub)

FUTURE

For the future, we may consider utilizing GPU accelerated libraries to handle complex models that contain smaller voxel size and therefore greater volume of voxels.

FUTURE



Matrix multiplication time spent using NumPy vs CuPy
Source: ArthurHoa (2022), GPU vs CPU (GitHub)

- Trimesh's overall performance is based on numpy and utilizes core linear algebra transforms like matrix multiplication.
- Matrix multiplication is O(n^3), but with GPU's matrix multiplication it leverages thousands of small, parallel cores to compute output elements simultaneously.
- Numpy offers a cuda fork called CuPa and utilizing GPU's will help speed the performance of our application, as shown in these graphs here. The HEART lab has GPU's to support the vision.

Changes to Requirements

- Favorite Magnetization
- Adding more materials
- Autosave to saving every action
- Property Validation (null)

QUESTIONS!!